INCH-POUND

MIL-H-0013514G(GL)
24 July 1991
USED IN LIEU OF
MIL-H-13514F
30 March 1988

MILITARY SPECIFICATION

HEATERS, SPACE, RADIANT TYPE, PORTABLE, 45,000 BTU/HR, (H-45)

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-H-13514F. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers 45,000 BTU/HR, H-45, portable radiant type space heaters (see 6.1).

1.2 Classification. Heaters shall be of the following types, as specified (see 6.2):

Type I - Solid fuel burning
Type II - Liquid fuel burning

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4520

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documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800	-	Fuel Oil, Diesel
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-P-551	-	Pipes and Elbows, Air Conditioning - Heating

STANDARDS

MILITARY

MIL-STD-105	-	Sampling procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-970	-	Standards and Specifications, Order of Preference for the Selection of
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks, are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-4400 - Heater, Space, Radiant, Large, Liquid Fuel, H-45, Type II
5-13-4415 - Burner, Shell Assembly
5-13-4437 - Hose Assembly, Flow Control - Burner
5-13-4460 - Heater, Space, Radiant, Large, Solid Fuel, H-45, Type I

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(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not specified shall be selected by the contractor in accordance with MIL-STD-970. Materials and components specified in accordance with MIL-STD-970 shall not degrade the operational suitability or effectiveness of the burner. It is encouraged that recycled material be used when practical providing the requirements of the specification are met.

3.2.1 Thread sealing compound. The thread sealing compound for the pipe joints shall be a commercial grade anti-seize compound (see 4.4.1.1).

3.2.2 Fuel. Fuel for testing the heater shall conform to grade DF-2 Continental U.S., Diesel conforming to VV-F-800.

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3.3 Design and construction. The design and construction of the type I and type II heaters shall conform to the following drawings, and as specified herein:

For type I	-	Drawing 5-13-4460 and all subsidiary drawings and parts lists pertaining thereto.
For type II	-	Drawing 5-13-4400 and all subsidiary drawings and parts lists pertaining thereto.

3.4 Type I and Type II heaters.

3.4.1 Leak test. Items 6 (up-tube), 7 (pipe connector), 8 (elbow), and 9 (pipe nipple) per Drawing 5-13-4415 and the hose assembly per Drawing 5-13-4437 shall be assembled and shall not leak when tested as specified in 4.4.2.1.

3.4.2 Combustion efficiency. When tested as specified in 4.4.5, the heater shall have a combustion efficiency greater than 50% when operated at 45,000 BTU/Hr or higher. The smoke reading shall not be greater than 4 on the Bacharach smoke scale during this test.

3.4.3 Smoke readings. The smoke reading of the heater shall not be greater than No. 4 on the Bacharach smoke scale when the heater is operating at 20,000 BTU/Hr or less or when the heater is operating at 45,000 BTU/Hr or greater when tested as specified in 4.4.5.

3.4.4 Combustion efficiency and Smoke reading after 20 hours of operation. When tested as specified in 4.5.5 the combustion efficiency and smoke reading requirement specified in 3.4.2 and 3.4.3 shall be met after the heater has been operating at 35,000 BTU/Hr for 20 consecutive hours.

3.5 Marking for identification. Heaters shall be marked for identification in accordance with the unit marking requirements of MIL-STD-130 and as specified on the applicable drawings.

3.6 Workmanship. The heaters shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

3.6.1 Surface defects. Space heaters shall be clean and free of distortion and misalignment of parts. All parts formed from sheet or strip steel shall be free of warpage, dents, wrinkles, bends not specified, fractures, sharp edges, burrs, and slivers. Casting and molded parts shall be free of sand, fins, pits, blow holes, and sprues. External surfaces shall be free from sharp edges.

3.6.2 Welds. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Spot welds shall be adherent and not burned through. Welds shall be continuous, sound, smooth, and free

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from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds, and rough edges and projections shall be dressed smooth.

3.6.3 Brazing. The surfaces of parts to be brazed shall be free from oxide, scale, paint, grease, and other foreign matter. Brazing shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed.

3.6.4 Threaded fittings. Threads shall not be missing, broken, stripped, or fractured. Threads shall be protected during welding, brazing, cleaning and painting from being distorted, damaged or coated. Thread sealing compound shall be applied to completely cover all threads and any excess after tightening shall be removed (see 3.2.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

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4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall consist of a sample size of the first 100 production units. The first article shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the thread sealing compound conforms to the requirement specified in 3.2.1.

4.4.2 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 Leak testing. Each burner assembly, consisting of assembled components specified in 3.4.1, shall have all openings sealed except the fuel inlet opening and the assembly shall be pressurized with air through the fuel intake opening at 5 pounds per square inch and examined for leaks. Examination may be made by immersion of pressurized assembly in a non-corrosive clear liquid and looking for bubbles or by measuring air pressure retention in the pressure assembly. Any decrease in pressure or the presence of a steady stream of bubbles eminating from the submerged assembly shall constitute failure of the test and shall be cause for the rejection of the burner assembly.

4.4.2.2 Visual examination of operations. Examination shall be made to determine if welding and brazing conforms to 3.6.2 and 3.6.3 and to drawings, application of sealing compound to threads conforms to 3.2.1 and 3.6.4, and

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fittings are protected from being distorted during brazing and welding in compliance with 3.6.4 and drawings. When nonconformance is noted, corrections shall be made to the items affected and to the operation. Items which cannot be corrected shall be removed from production.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Welding	Any weld missing, incomplete cracked, fractured, has burn holes or fusion incomplete	101	
	Not smooth, scale or flux deposits, excess metal not removed		201
	Blowholes, hard spots, cold shuts, or porosity	102	
	Not free from spatter		202
Brazing	Any braze missing, not smooth or not continuous	103	
	Flux not removed		203
Threaded components	Threads broken, stripped, fractured or missing	104	
	Components not securely tightened		204
	Excess sealing compound not removed		205
Design, construction and workmanship (applicable to all components)	Incorrect type of heater	105	
	Any component missing or not as specified	106	
	Sheet metal fractured, dented, torn, wrinkled, or otherwise malformed	107	
	Any component misplaced or not in proper alignment		206
	Sharp edges, burrs, or slivers		207
	Any loose particles inside tank (as noted when shaking)		208

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Design, construction and workmanship (applicable to all components) (cont'd)	Any operation omitted or not performed as specified	108	
	Any functioning component that is inoperative or will not operate as intended	109	
	Castings with prominent parting lines due to improper molding	110	
	Cracks	111	
	Not clean		209
	Projections or metal breakout from mold	112	
Heater assembly	Collar improperly formed or trimmed, top edge of uneven height, sharp or turned outward, or loose	113	
	Spot welds broken, missing, or cratered due to metal burning	114	
	Sloppy out of tolerance fit between heater top and bottom shells and burner adapter ring	115	
	Lid distorted or opened around flanged edge		210
	Spot welds on handle less than two	116	
	One or both doors missing	117	
	Door does not fit contour of front base (opening between door and base)	118	
	Hinge not in proper location	119	
	Door latch embossing missing or too small		211
	Door latch not free turning		212
	Door latch rivet improperly peened or not peened		213
	Door out of alignment		214
	Hinge binds		215
	High fire ring not securely positioned in its collar, or high fire ring obstructs holes in top of shell	120	
	Retaining loop, eye bolt, or wing nut missing or improperly located	121	
	Instruction plate dirty, or lettering smudged		216

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Heater assembly (cont'd)	Retainer loop binds (does not swing freely of its own weight)		217
	Guy line cord cut or abraded	122	
	Guy line cord not solid braided, or not tied with double overhand knot		218
	Guy line cord is glazed, excessively greasy, tacky, oily, or stiff		219
	Eyebolts not peened or improperly peened		220
	Valve bracket support missing or improperly located	123	
	Improper height of fuel valve in relation to burner shell assembly and burner up-tube	124	
	Grate door assembly missing or improperly located	125	
	Cup/cable assembly missing	126	
Grates	Warped more than 1/8 inch in any direction	127	
Marking for identification	Missing, illegible, incomplete, incorrect	128	
	Wrong size, or wrong location		221

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items shall be tested for the characteristics listed in table III. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be S-1. Any test failure shall be cause for the rejection of the lot.

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TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Combustion efficiency	3.4.2	4.5.3
Smoke reading	3.4.3	4.5.4
Combustion efficiency and smoke reading after 20 hours of operation	3.4.4	4.5.5

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size location, sequence, or method of application
Preservation	Improperly applied or missing
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.5 Methods of inspection.

4.5.1 Heater test setup. For all heater tests the following minimum conditions apply:

a. The assembled heater shall have no less than six lengths of stove pipe and the flue cap. The heater assembly shall be level.

b. Drill two holes (approximately 1/4" dia.) in the second stack section, 36 inches ± 1 inch, above the heater top surface where all stack temperatures, carbon dioxide (CO_2), smoke and draft measurements, (see 6.3) shall be taken. The ambient temperature

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next to the area of the fuel inlet shall be recorded. The draft shall be natural in flow without the use of any mechanical draft-inducer.

c. A thermometer capable of determining a temperature up to 1300°F is necessary to determine the stack temperature.

d. A Bacharach smoke tester/scale shall be used. A 100cc soap film flow meter shall be used to measure the gravity fed fuel flow input rate to the heater.

e. An interval of not less than 15 minutes shall be maintained between each change in the firing rate to allow for flame stabilization. Final smoke number determinations shall be the average of three consecutive readings.

4.5.3 Combustion efficiency test. The heater shall be operated until a heater out put of 45,000 BTU/hr or higher is attained and the combustion efficiency shall then be determined using the following formula:

$$E = 1 - \left[\frac{3171 + 2.69\ TS + \left[\frac{216.2 + 0.77}{CO_2} \right] (TS - TI) - 5.38\ TI}{HHV} \right]$$

E = Combustion efficiency (%)
TS = Stack temperature (degrees C) (see 4.5.1b)
TI = Inlet ambient temperature (degrees C) (see 4.5.1b)
CO_2 = Carbon dioxide (%) (see 4.5.1b)
HHV = High heat value (joules/gram)

A test failure shall be recorded if the Combustion efficiency is not more than 50% and the smoke reading is greater than 4 on the Bacharach smoke scale.

4.5.4 Smoke reading test.

4.5.4.1 Smoke reading low fire. The heater shall be operated until a heater out put of not more than 20,000 BTU/Hr is attained. Then three consecutive smoke readings using the Bacharach smoke scale shall be determined while the heater is operated at no more than 20,000 BTU/Hr. and the average of the readings recorded. An average smoke reading of greater than 4 shall be recorded as a test failure. Heater output shall be determined as follows:

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Heater output (BTU/hr) = F X P X HHV/1055 X E X 60

F = Fuel flow rate (ml/min)
P = Fuel density is 0.852 gm/ml for DF-2 fuel
HHV = High heat value of DF-2 fuel is 45166 j/gm
E = % Combustion Efficiency attained using formula in 4.5.3, and the TS, TI, and $\%CO_2$ values as determined above.

4.5.4.2 Smoke reading high fire. The heater shall be operated until an out put of 45,000 BTU/hr BTU/Hr or greater is attained. The smoke reading shall be determined according to the procedures indicated in 4.5.4.1. A smoke reading greater than 4 on the Bacharach smoke scale shall be recorded as a test failure.

4.5.5 Smoke reading and Combustion efficiency after 20 hours of operation test. The heater shall be operated for 20 consecutive hours at 35,000 BTU/Hr and tested as specified in 4.5.3 and 4.5.4.1. A test failure shall be recorded if the Combustion efficiency is not more than 50% and the smoke reading is greater than 4 on the Bacharach smoke scale.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Preservative application.

5.1.1.1.1 Type I and type II heaters. All uncoated ferrous metal surfaces of each heater and components, except the pipe sections, shall be cleaned by method C-1, thoroughly dried and coated with type F-1 preservative in accordance with MIL-P-116. The pipe sections shall be coated with preservative in accordance with MIL-P-551. All components not otherwise specified shall be preserved in accordance with method III of MIL-P-116.

5.1.1.2 Unit packing.

5.1.1.2.1 Pipe sections. Six sections of pipe, preserved as specified in 5.1.1.1.1, shall be unit packed in accordance with MIL-P-551, except that prior to wrapping the pipe sections for the type I heater, one poker, one shaker, and one damper, preserved as specified in 5.1.1.1.1, shall be placed in the center of the pipe sections. The size of the fiberboard box shall not exceed outside dimensions of 25 inches by 6 inches by 6 inches.

5.1.2 Commercial preservation. The pipe sections, poker, shaker, and damper shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Type I heater. One complete type I heater shall be packed into a close-fitting cleated-plywood box conforming to style A or I of PPP-B-601. The inside dimensions of the shipping container shall be approximately 22 inches in length, 22 inches in width, and 27 inches in depth. The heater shall be packed in an upright position with all components (see Drawing 5-13-4460), except the pipe, placed inside the heater with top and bottom shells locked together. The pipe shall be placed next to the heater in the shipping container. The components placed inside the heater shall be cushioned, blocked and braced and rendered immobile in accordance with MIL-STD-1186, so that no damage will occur with vibration. The whole heater assembly shall be secured in the shipping container. The shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

5.2.1.2 Type II heater. One complete type II heater shall be packed by the same requirements as specified in 5.2.1.1, except that the components shall be packed in accordance with Drawing 5-13-4400.

5.2.2 Level B packing.

5.2.2.1 Type I heater. One complete type I heater shall be packed as specified in 5.2.1.1, except the shipping container shall conform to a grade V13c, class weather-resistant, corrugated fiberboard container in accordance with PPP-B-636. The container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

5.2.2.2 Type II heater. One complete type II heater shall be packed as specified in 5.2.1.2, except the shipping container shall conform to a grade V13c, class weather-resistant, corrugated fiberboard container in accordance with the appendix of PPP-B-636. The container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

5.2.3 Commercial packing. The type I and II complete heaters, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special markings required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The space heaters are intended to heat tents and temporary shelters.

6.1.1 Type I. The type I space heater is intended for use with coal or other solid fuel.

6.1.2 Type II. The type II space heater is intended for use with all liquid fuels (JP-4, JP-5, JP-8, kerosene, DF-A, DF-1, DF-2, and gasoline).

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of space heater required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection and approval of the first article.

6.4 Bacharach Combustion Kit. A Bacharach Combustion Kit has been found to be suitable for testing carbon dioxide (CO_2) and for determining smoke values. The Bacharach Combustion Kit may be acquired from Bacharach Inc. 625 Alpha Drive, Pittsburgh, PA 15238, (412) 963-2000.

6.5 Subject term (keyword) listing.

Coal burning
DF-A; DF-1; DF-2
Gasoline
JP-4; JP-5; JP-8
Kerosene
Liquid fuel burning
Solid fuel burning

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Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - ME
Air Force - 84
DLA - CS

User activities:

Army - CE
Navy - MC

Preparing activity:

Army - GL

(Project 4520-A010)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-0013514G(GL)	1991 July 24

3. DOCUMENT TITLE

HEATERS, SPACE, RADIANT TYPE, PORTABLE, 45,000 BTU/HR, (H-45)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4531	256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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